Healthcare Data Analysis & Preprocessing

Midterm Project Proposal | Exploratory Data Analysis and Machine Learning Preparation

Dataset Overview

This project uses the professor-approved Healthcare Dataset published on Kaggle by prasad22. The original CSV contains 55,500 records and 15 columns covering patient demographics, medical conditions, admission and discharge dates, insurance, billing amounts, and test results. The dataset is synthetic and is intended for educational analysis, not clinical decision-making.

Project Objective

The objective is to examine healthcare-related patterns and develop a reproducible preprocessing workflow that prepares training data for a future machine learning study. One possible future task is to explore classification of the dataset's Test Results categories (Normal, Abnormal, and Inconclusive), while checking whether available predictors provide meaningful predictive information.

Methods and Workflow

Using Python in JupyterLab, the dataset is split into 70% training (38,850 records) and 30% testing (16,650 records). Exploratory data analysis and preprocessing are performed on training data only. Initial checks identified no missing values and 251 exact duplicate training records, which were removed. A Length of Stay feature was derived from admission and discharge dates. Five categorical features were one-hot encoded, and Age, Billing Amount, and Length of Stay were scaled using Min-Max scaling fitted only on training data. Matplotlib charts document distributions and before-and-after scaling.

Preliminary Findings and Next Steps

The exported processed training file contains 38,599 records and 31 columns. Test Results categories appear approximately balanced, so resampling was not indicated by the initial class counts. Some original text fields and identifiers remain in the export; these must be assessed and appropriately excluded or transformed before training a model. Future work would define the prediction target, screen for leakage and unsuitable identifiers, build a consistent feature pipeline, and evaluate models on held-out data without fitting preprocessing steps on the test set.